Continuity

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Testing continuity**

Consider the piecewise function $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$.

a Find $\lim_{x \to 2^-} f(x)$. $= 5$.

b Find $\lim_{x \to 2^+} f(x)$. $= 5$.

c Is f continuous at $x = 2$? Justify. Yes — both one-sided limits equal $5 = f(2)$, so f is continuous at $x = 2$.

(a) $= 5$.; (b) $= 5$.; (c) Yes — both one-sided limits equal $5 = f(2)$, so f is continuous at $x = 2$.

2 **[Foundational]** • **Target: Testing continuity**

The piecewise function $f(x) = kx + 1$ for $x \leq 3$, and $f(x) = x^2 - 2$ for $x > 3$, is continuous at $x = 3$. Find the value of k.

Need $3k + 1 = 9 - 2 = 7$, so $k = 2$.

3 **[Foundational]** • **Target: Types of discontinuity and the Intermediate Value Theorem**

Classify the discontinuity of each function at the given point:

a $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$ Removable (a hole) — the factor $(x - 1)$ cancels.

b $f(x) = \frac{1}{x - 2}$ at $x = 2$ Infinite discontinuity — f blows up as $x \to 2$.

c $f(x) = \lfloor x \rfloor$ at $x = 3$ Jump discontinuity — the floor function jumps from 2 to 3.

(a) Removable (a hole) — the factor $(x - 1)$ cancels.; (b) Infinite discontinuity — f blows up as $x \to 2$.; (c) Jump discontinuity — the floor function jumps from 2 to 3.

4 **[Foundational]** • **Target: Types of discontinuity and the Intermediate Value Theorem**

Show that $f(x) = x^3 - 4x + 1$ has a root in the interval $(1, 2)$, using the Intermediate Value Theorem.

$f(1) = 1 - 4 + 1 = -2 < 0$ and $f(2) = 8 - 8 + 1 = 1 > 0$. Since f is continuous (a polynomial) and changes sign, the IVT guarantees a root in $(1, 2)$.

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5 [Foundational] • Target: Continuity on an interval

State the largest interval(s) on which each function is continuous:

a $f(x) = \frac{1}{x-4}$ Continuous everywhere except $x = 4$: on $(-\infty, 4) \cup (4, \infty)$.

b $f(x) = \sqrt{x-3}$ Continuous on $[3, \infty)$.

c $f(x) = \ln(x+2)$ Continuous on $(-2, \infty)$.

(a) Continuous everywhere except $x = 4$: on $(-\infty, 4) \cup (4, \infty)$.; (b) Continuous on $[3, \infty)$.; (c) Continuous on $(-2, \infty)$.

6 [Foundational] • Target: Continuity on an interval

Find the value of c that makes $f(x) = \frac{x^2-9}{x-3}$ for $x \neq 3$, and $f(3) = c$, continuous at $x = 3$.

$\lim_{x \to 3} \frac{x^2-9}{x-3} = \lim_{x \to 3} (x+3) = 6$, so $c = 6$.

7 [Foundational] • Target: Applying the Intermediate Value Theorem

Show that $f(x) = \cos x - x$ has a root in the interval $(0, 1)$.

$f(0) = \cos 0 - 0 = 1 > 0$ and $f(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.460 < 0$. Since f is continuous and changes sign on $[0, 1]$, IVT guarantees a root in $(0, 1)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Applying the Intermediate Value Theorem

A hiker starts a trail at elevation 200 m at 8:00 AM and finishes at elevation 800 m at noon. Explain, using the IVT, why the hiker must have been at elevation exactly 500 m at some point during the hike.

Elevation as a function of time is continuous (the hiker moves without teleporting), starts below 500 m and ends above 500 m, so by the IVT the hiker's elevation must equal 500 m at some intermediate time.

9 [Intermediate] • Target: Visualizing discontinuities

The function $f(x) = \frac{1}{x-2}$ (with an infinite discontinuity at $x = 2$) is graphed below. Use the graph to describe the one-sided behavior near $x = 2$.

As $x \to 2^-$, $f(x) \to -\infty$; as $x \to 2^+$, $f(x) \to +\infty$ — the function is unbounded on both sides near $x = 2$, confirming an infinite (non-removable) discontinuity.

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10 [Intermediate] • Target: Continuity of composite and combined functions

Explain why the composition of two continuous functions is continuous, and use this to argue that $h(x) = \sqrt{x^2 + 1}$ is continuous everywhere.

If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a (limits pass through continuous functions). Here $x^2 + 1$ is a continuous polynomial (always $\geq 1 > 0$), and $\sqrt{u}$ is continuous for $u \geq 0$, so the composition $\sqrt{x^2 + 1}$ is continuous for all real x.

11 [Intermediate] • Target: Removable discontinuities and redefinition

The function $f(x) = \frac{x^2 + x - 6}{x - 2}$ has a removable discontinuity at $x = 2$. Find the value that, if assigned to $f(2)$, would make f continuous there.

Simplify: $\frac{(x-2)(x+3)}{x-2} = x + 3$ for $x \neq 2$. $\lim_{x \to 2} f(x) = 2 + 3 = 5$. Assigning $f(2) = 5$ removes the discontinuity.

12 [Intermediate] • Target: Removable discontinuities and redefinition

Find all values of x where $f(x) = \frac{x^2 - 16}{x^2 - 2x - 8}$ has a discontinuity, and classify each as removable or non-removable.

Factor: $\frac{(x-4)(x+4)}{(x-4)(x+2)}$. Common factor $(x - 4)$: removable discontinuity at $x = 4$. Remaining denominator zero at $x = -2$: non-removable (infinite) discontinuity.

13 [Intermediate] • Target: The Extreme Value Theorem (a continuity consequence)

State the Extreme Value Theorem, and explain why $f(x) = x^2$ is guaranteed to have an absolute maximum and minimum on $[-3, 2]$.

The EVT states a continuous function on a closed interval $[a, b]$ attains both an absolute max and min on that interval. Since $f(x) = x^2$ is continuous (a polynomial) and $[-3, 2]$ is closed, the EVT guarantees both extrema exist (here: min at $x = 0$, max at $x = -3$).

14 [Intermediate] • Target: Types of discontinuity and the Intermediate Value Theorem

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{3^{-n}(x-3)^n}{n}$.

The radius is 3 (ratio test). At $x = 6$ the series is $\sum_{n=1}^{\infty} \frac{1}{n}$; at $x = 0$ it is $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$. Checking each endpoint gives $[0, 6)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Continuity on an interval • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Find the directional derivative of $f(x, y) = x\cos(y) - y$ at $(-2, \ -2)$ in the direction of $\langle -3, 4 \rangle$.

$ablaf(-2, \ -2) = \langle \cos(2), \ -2\sin(2) - 1 \rangle$ and the unit vector is $\langle -\frac{3}{5}, \frac{4}{5} \rangle$.
$D_{\mathbf{u}} f = ablaf \cdot \mathbf{u} = -\frac{8\sin(2)}{5} - \frac{4}{5} - \frac{3\cos(2)}{5}$.

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16 [Challenging] • Target: Applying the Intermediate Value Theorem • Cross-Lesson: 11.01 Sequences and Series

If $z = \ln(x^2 + y^2)$ with $x = t^2$ and $y = \sin(t)$, find $\frac{dz}{dt}$ at $t = 1$.

$\frac{dz}{dt} = f_x\frac{dx}{dt} + f_y\frac{dy}{dt}$. At $t = 1$: $\frac{2}{\sin^2(1)+1} \cdot 2 + \frac{2\sin(1)}{\sin^2(1)+1} \cdot \cos(1) = \frac{2(\sin(2)+4)}{3-\cos(2)}$.

17 [Challenging] • Target: Visualizing discontinuities • Cross-Lesson: 12.01 Convergence Tests

Evaluate $\int_0^{\pi} e^{3x}\cos(2x)\,dx$.

Integrate by parts twice; the original integral reappears, so solve for it:
$\int e^{3x}\cos(2x)\,dx = \frac{(2\sin(2x)+3\cos(2x))e^{3x}}{13}$. From 0 to π: $-\frac{3}{13} + \frac{3e^{3\pi}}{13}$.

18 [Challenging] • Target: Continuity of composite and combined functions • Cross-Lesson: 12.02 Power Series and Taylor Series

Flux of $\mathbf{F} = (x, y, 0)$ through lateral surface of cylinder $x^2 + y^2 = 1$, $0 \le z \le 2$, outward:

$abla \cdot \mathbf{F} = 2$; by divergence theorem on cylinder: $2 \cdot \text{Vol} = 2 \cdot 2\pi - \text{caps}$. Easier: direct:
$\hat{n} = (\cos\theta, \sin\theta, 0)$, $\mathbf{F} = (\cos\theta, \sin\theta, 0) \Rightarrow \mathbf{F} \cdot \hat{n} = 1$; $\iint_S 1\,dS = 2\pi \cdot 2 = 4\pi$.

19 [Challenging] • Target: Removable discontinuities and redefinition • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Evaluate $\int_0^{\pi} e^{3x}\sin(2x)\,dx$.

Integrate by parts twice; the original integral reappears, so solve for it:
$\int e^{3x}\sin(2x)\,dx = \frac{(3\sin(2x)-2\cos(2x))e^{3x}}{13}$. From 0 to π: $\frac{2}{13} - \frac{2e^{3\pi}}{13}$.

20 [Challenging] • Target: The Extreme Value Theorem (a continuity consequence) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Evaluate $\int_0^{\pi} e^{-x}\cos(2x)\,dx$.

Integrate by parts twice; the original integral reappears, so solve for it:
$\int e^{-x}\cos(2x)\,dx = \frac{(2\sin(2x)-\cos(2x))e^{-x}}{5}$. From 0 to π: $-\frac{1-e^{\pi}}{5e^{\pi}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Definition of Continuity at a Point

Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ **is defined,** $\lim_{x \to c} f(x)$ **exists, and** $\lim_{x \to c} f(x) = f(c)$ ✓

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

d. $f(c) = 0$ and $f'(c) = 0$

The formal 3-part definition of continuity at a point requires $f(c)$ is defined, the two-sided limit exists, and the limit equals the function value.

22 [Foundational] • Target: Definition of Continuity at a Point

Determine if $f(x)$, defined as $\frac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $f(2) = 5$, is continuous at $x = 2$.

a. **Discontinuous (removable discontinuity at** $x = 2$**)** ✓

b. Continuous at $x = 2$

c. Discontinuous (jump discontinuity)

d. Discontinuous (infinite discontinuity)

$\lim_{x \to 2} \frac{(x - 2)(x + 2)}{x - 2} = 4$. However, $f(2) = 5 \neq 4$. The limit exists but differs from $f(2)$, so it is a removable discontinuity.

23 [Foundational] • Target: Definition of Continuity at a Point

For what value of a is $h(x)$, defined as $ax^2 + 1$ for $x \leq 2$ and $3x - a$ for $x > 2$, continuous everywhere?

a. 1 ✓

b. 2

c. $3/5$

d. -1

Continuity at $x = 2$: $a(2)^2 + 1 = 3(2) - a \Rightarrow 4a + 1 = 6 - a \Rightarrow 5a = 5 \Rightarrow a = 1$.

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24 [Foundational] • Target: Classification of Discontinuities

Classify the discontinuity of $g(x) = \frac{x-1}{|x-1|}$ at $x = 1$.

a. Jump discontinuity ✓

b. Removable discontinuity

c. Infinite discontinuity

d. Essential oscillating discontinuity

$\lim_{x \to 1^+} g(x) = 1$ while $\lim_{x \to 1^-} g(x) = -1$. Because both one-sided limits exist but are unequal, it is a jump discontinuity.

25 [Foundational] • Target: Classification of Discontinuities

Classify the discontinuity of $f(x) = \frac{1}{(x-4)^2}$ at $x = 4$.

a. Infinite discontinuity ✓

b. Removable discontinuity

c. Jump discontinuity

d. Continuous

As $x \to 4$, $f(x) \to +\infty$. The function grows without bound, which defines an infinite discontinuity (vertical asymptote).

26 [Foundational] • Target: Classification of Discontinuities

How can $f(x) = \frac{\sin(3x)}{x}$ be defined at $x = 0$ to make it continuous?

a. Set $f(0) = 3$ ✓

b. Set $f(0) = 1$

c. Set $f(0) = 0$

d. It cannot be made continuous

Since $\lim_{x \to 0} \frac{\sin(3x)}{x} = 3$, setting $f(0) = 3$ removes the discontinuity, making f continuous.

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27 [Foundational] • Target: Intermediate Value Theorem (IVT)

State the primary hypothesis required to apply the Intermediate Value Theorem to f on $[a, b]$.

a. f **must be continuous on the closed interval** $[a, b]$ ✓

b. f must be differentiable on (a, b)

c. $f(a) = f(b)$

d. $f'(x) > 0$ on $[a, b]$

The IVT states that if f is continuous on $[a, b]$ and d is between $f(a)$ and $f(b)$, there exists at least one $c \in (a, b)$ such that $f(c) = d$.

28 [Foundational] • Target: Intermediate Value Theorem (IVT)

Use the IVT to show that $x^3 + 2x - 5 = 0$ has a root in which interval?

a. $(1, 2)$ ✓

b. $(0, 1)$

c. $(2, 3)$

d. $(-1, 0)$

Let $f(x) = x^3 + 2x - 5$. $f(1) = 1 + 2 - 5 = -2 < 0$. $f(2) = 8 + 4 - 5 = 7 > 0$. Since f is continuous and changes sign, a root exists in $(1, 2)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Intermediate Value Theorem (IVT)

A continuous function g satisfies $g(0) = -3$ and $g(4) = 5$. Does $g(x) = \pi$ have a solution on $(0, 4)$?

a. **Yes, because** $-3 < \pi < 5$ **and** g **is continuous** ✓

b. No, because π is irrational

c. Only if g is differentiable

d. Cannot be determined without $g'(x)$

Since $-3 < \pi \approx 3.1416 < 5$ and g is continuous on $[0, 4]$, the IVT guarantees at least one $c \in (0, 4)$ with $g(c) = \pi$.

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30 [Intermediate] • Target: Extreme Value Theorem (EVT) and Properties

What does the Extreme Value Theorem guarantee for a function continuous on a closed bounded interval $[a, b]$?

a. f **attains both an absolute maximum and an absolute minimum on** $[a, b]$ ✓

b. f has at least one inflection point

c. f is invertible on $[a, b]$

d. $f'(c) = 0$ for some $c \in (a, b)$

The EVT guarantees that any continuous function on a closed bounded interval $[a, b]$ attains both an absolute maximum and an absolute minimum.

31 [Intermediate] • Target: Extreme Value Theorem (EVT) and Properties

Which function fails to achieve an absolute maximum on $(0, 1)$?

a. $f(x) = \frac{1}{x}$ ✓

b. $f(x) = x^2$

c. Both $f(x) = 1/x$ and $f(x) = x^2$

d. $f(x) = 5$

On open interval $(0, 1)$, $1/x \to \infty$ (unbounded) and $x^2 < 1$ never reaches 1. Both fail the EVT because the interval is open, not closed.

32 [Intermediate] • Target: Extreme Value Theorem (EVT) and Properties

If f and g are continuous at $x = c$, which of the following is NOT guaranteed to be continuous at $x = c$?

a. $f(x)/g(x)$ ✓

b. $f(x) + g(x)$

c. $f(x) \cdot g(x)$

d. $[f(x)]^2$

The quotient f/g is only continuous at c if $g(c) \neq 0$. If $g(c) = 0$, f/g is undefined and discontinuous.

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33 [Intermediate] • Target: Testing continuity

A continuous function on a closed, bounded interval $[a, b]$ is guaranteed (Extreme Value Theorem) to be:

a. bounded and to attain its max and min ✓

b. monotonic

c. unbounded

d. differentiable

The Extreme Value Theorem: continuity on a compact interval guarantees both boundedness and attained extrema.

34 [Intermediate] • Target: Types of discontinuity and the Intermediate Value Theorem

Removable discontinuity: $f(x) = \frac{x^2-4}{x-2}$. To make f continuous at $x = 2$, define $f(2) =$

a. 4 ✓

b. 2

c. 0

d. -4

Applying Limits and Continuity principles, $(x^2 - 4)/(x - 2) = x + 2 \to 4$. Therefore, the verified correct option is 4.

35 [Intermediate] • Target: Continuity on an interval

By the Intermediate Value Theorem, does $f(x) = x^3 - 4x - 2$ have a root in $[2, 3]$?

a. No $-$ $-$ $f(2)$ and $f(3)$ have the same sign

b. Yes $-$ $-$ $f(2)$ and $f(3)$ have opposite signs ✓

c. No $-$ $-$ f is not continuous on $[2, 3]$

d. Cannot be determined without more information

$f(2) \approx -2.00$ and $f(3) \approx 13.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(2, 3)$.

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36 [Intermediate] • Target: Applying the Intermediate Value Theorem

By the Intermediate Value Theorem, does $f(x) = x^5 - x - 1$ have a root in $[1, 2]$?

a. No — $f(1)$ and $f(2)$ have the same sign

b. Cannot be determined without more information

c. Yes — $f(1)$ and $f(2)$ have opposite signs ✓

d. No — f is not continuous on $[1, 2]$

$f(1) \approx -1.00$ and $f(2) \approx 29.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

37 [Intermediate] • Target: Visualizing discontinuities

By the Intermediate Value Theorem, does $f(x) = x^4 - x - 1$ have a root in $[1, 2]$?

a. Yes — $f(1)$ and $f(2)$ have opposite signs ✓

b. No — $f(1)$ and $f(2)$ have the same sign

c. Cannot be determined without more information

d. No — f is not continuous on $[1, 2]$

$f(1) \approx -1.00$ and $f(2) \approx 13.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Continuity of composite and combined functions • Cross-Lesson: 12.02 Power Series and Taylor Series

By the Intermediate Value Theorem, does $f(x) = x^3 - 2x - 5$ have a root in $[2, 3]$?

a. Yes — $f(2)$ and $f(3)$ have opposite signs ✓

b. Cannot be determined without more information

c. No — f is not continuous on $[2, 3]$

d. No — $f(2)$ and $f(3)$ have the same sign

$f(2) \approx -1.00$ and $f(3) \approx 16.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(2, 3)$.

39 [Challenging] • Target: Removable discontinuities and redefinition • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

By the Intermediate Value Theorem, does $f(x) = x^3 + 4x^2 - 10$ have a root in $[1, 2]$?

a. No — f is not continuous on $[1, 2]$

b. Cannot be determined without more information

c. Yes — $f(1)$ and $f(2)$ have opposite signs ✓

d. No — $f(1)$ and $f(2)$ have the same sign

$f(1) \approx -5.00$ and $f(2) \approx 14.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

40 [Challenging] • Target: The Extreme Value Theorem (a continuity consequence) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

By the Intermediate Value Theorem, does $f(x) = x^3 - 6x + 2$ have a root in $[2, 3]$?

a. Cannot be determined without more information

b. No — f is not continuous on $[2, 3]$

c. Yes — $f(2)$ and $f(3)$ have opposite signs ✓

d. No — $f(2)$ and $f(3)$ have the same sign

$f(2) \approx -2.00$ and $f(3) \approx 11.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(2, 3)$.

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41 [Challenging] • Target: Definition of Continuity at a Point • Cross-Lesson: 15.01 Double Integrals

By the Intermediate Value Theorem, does $f(x) = x^5 - 3x - 1$ have a root in $[1, 2]$?

a. No — $f(1)$ and $f(2)$ have the same sign

b. No — f is not continuous on $[1, 2]$

c. Cannot be determined without more information

d. Yes — $f(1)$ and $f(2)$ have opposite signs ✓

$f(1) \approx -3.00$ and $f(2) \approx 25.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

42 [Challenging] • Target: Classification of Discontinuities • Cross-Lesson: 16.01 Vector Fields and Line Integrals

By the Intermediate Value Theorem, does $f(x) = x^3 - 5x - 2$ have a root in $[2, 3]$?

a. Cannot be determined without more information

b. No — $f(2)$ and $f(3)$ have the same sign

c. No — f is not continuous on $[2, 3]$

d. Yes — $f(2)$ and $f(3)$ have opposite signs ✓

$f(2) \approx -4.00$ and $f(3) \approx 10.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(2, 3)$.

43 [Challenging] • Target: Intermediate Value Theorem (IVT) • Cross-Lesson: 2.01 Evaluating Limits

By the Intermediate Value Theorem, does $f(x) = x^5 - 4x - 1$ have a root in $[1, 2]$?

a. No — f is not continuous on $[1, 2]$

b. Cannot be determined without more information

c. Yes — $f(1)$ and $f(2)$ have opposite signs ✓

d. No — $f(1)$ and $f(2)$ have the same sign

$f(1) \approx -4.00$ and $f(2) \approx 23.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

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44 [Challenging] • Target: Extreme Value Theorem (EVT) and Properties • Cross-Lesson: 3.01 The Derivative from First Principles

By the Intermediate Value Theorem, does $f(x) = x^5 - 2x - 3$ have a root in $[1, 2]$?

a. Cannot be determined without more information

b. No — $f(1)$ and $f(2)$ have the same sign

c. No — f is not continuous on $[1, 2]$

d. Yes — $f(1)$ and $f(2)$ have opposite signs ✓

$f(1) \approx -4.00$ and $f(2) \approx 25.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

45 [Challenging] • Target: Testing continuity • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

By the Intermediate Value Theorem, does $f(x) = x^3 + 4x - 15$ have a root in $[1, 2]$?

a. Cannot be determined without more information

b. No — $f(1)$ and $f(2)$ have the same sign

c. No — f is not continuous on $[1, 2]$

d. Yes — $f(1)$ and $f(2)$ have opposite signs ✓

$f(1) \approx -10.00$ and $f(2) \approx 1.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

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